

Use-Case Flow Specification

Name: MANASA K

SRN:PES1UG25CS824

Use Case ID	UC-05
Use Case Name	Reserve Food Batch
Primary Actor	Shelter Coordinator
Secondary Actor(s)	Donor Partner (notified of reservation), System (generates verification code)
Trigger	Shelter Coordinator selects “Reserve” on a food batch listed on the claim board.

Preconditions

- The Shelter Coordinator is authenticated and verified on the platform.
- At least one food batch is listed on the claim board, is unreserved, and its expiry countdown timer has not reached zero.
- The Shelter Coordinator has viewed the batch details (quantity, dietary tags, expiry time, donor location).

Postconditions

- Success: The batch status changes to “Reserved” and is removed from other shelters’ browse/claim view; a unique pickup verification code and pickup window are generated and shared with the Shelter Coordinator; the Donor Partner is notified of the reservation.
- Failure: The batch remains “Available” on the claim board and no verification code is generated.

Main Success Scenario

1. Shelter Coordinator browses/filters the claim board and selects a desired food batch to view its details.
2. Shelter Coordinator taps “Reserve Batch.” System checks that the batch is still available and unexpired.
3. System locks the batch, changes its status to “Reserved,” and removes it from other shelters’ view.
4. System generates a unique pickup verification code and a suggested pickup time window.
5. System displays the verification code and pickup window to the Shelter Coordinator and sends a confirmation notification to the Donor Partner.
6. Shelter Coordinator travels to the donor location and presents the verification code at pickup; Donor Partner confirms handoff. Use case ends successfully.

Alternate Flow – Batch Reserved by Another Shelter (Race Condition)

1. 2a. At Step 2, the system detects that the batch was reserved by another Shelter Coordinator a moment earlier.
2. 2a1. System displays a “Batch No Longer Available” message and removes the batch from this coordinator’s view.
3. 2a2. System suggests similar available batches matching the coordinator’s dietary/distance filters, if any exist.
4. 2a3. Shelter Coordinator selects an alternative batch and the flow resumes at Step 1, or exits the flow. Use case ends without reservation.